

Spanning Tree Protocol (STP) Topology Loop Predictor

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Abstract & Introduction

Abstract

The STP Topology Loop Predictor is a software tool that analyzes Layer-2 network configurations, builds the network topology, simulates STP behavior, detects loop risks, and validates security features such as BPDU Guard and Loop Guard. It helps network administrators identify problems before deployment.

Introduction

Layer-2 networks use redundant links for reliability. However, these links may create switching loops, causing broadcast storms. STP prevents loops by blocking unnecessary paths. This project automates STP analysis and predicts potential network failures.

Problem Statement & Objectives

Problem Statement

- Loop creation due to redundant links
- STP misconfiguration
- Rogue Root Bridge attacks
- Unidirectional link failures

Objectives

- Build Layer-2 topology
- Simulate STP
- Detect loop risks
- Validate security settings
- Generate AI recommendations

Existing & Proposed System

Existing System	Proposed System
Manual analysis using CLI commands such as <code>show spanning-tree</code> .	Automatically analyzes uploaded switch configuration files.
Requires network administrators to inspect configurations manually.	Builds the Layer-2 topology automatically using graph algorithms.
Uses simulators like GNS3 and EVE-NG, which consume high system resources.	Lightweight Python-based simulator with fast processing.
Cannot proactively predict network loops before deployment.	Predicts Layer-2 loops by simulating STP/RSTP behavior.
Limited security validation for STP features.	Validates BPDU Guard, Root Guard, Loop Guard, and UDLD configurations.
Troubleshooting is time-consuming and error-prone.	Detects configuration issues and provides AI-based recommendations instantly.
Difficult to scale for large enterprise networks.	Supports large-scale networks with efficient graph-based analysis.
Report generation is mostly manual.	Automatically generates loop risk reports and configuration suggestions

System Architecture & Workflow

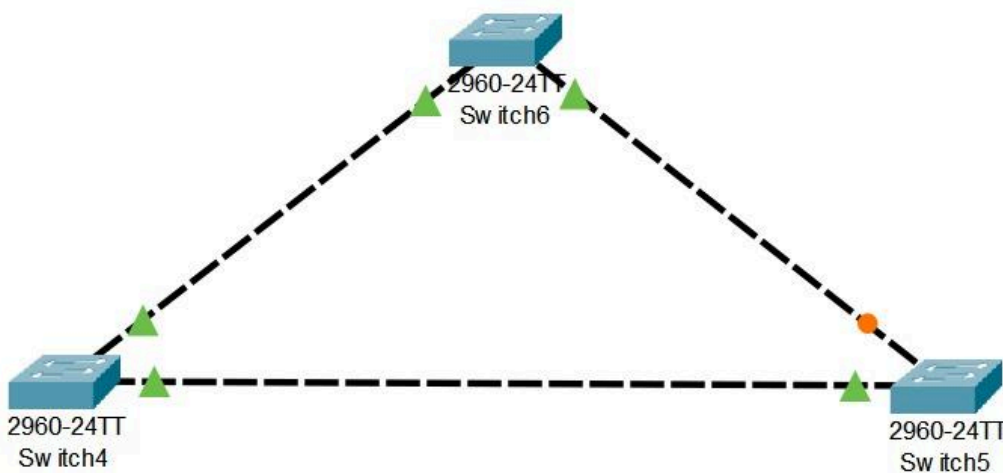
Architecture

Modules:

- Configuration Upload
- Configuration Parser
- Topology Builder
- STP Simulation
- Loop Analyzer

- AI Recommendation
- Report Generator

CISCO PACKET TRACER MODEL



Technology Stack

Component	Technology Used
Frontend	React.js
Backend	FastAPI
Database	PostgreSQL
Graph Processing	NetworkX
User Interface (UI)	Tailwind CSS
AI Integration	Gemini API

Conclusion

Advantages

- Automatic topology mapping
- STP simulation
- Early loop detection
- Security validation
- AI-powered recommendations

Future Scope

- Live network monitoring
- SNMP integration
- Multi-vendor support
- MSTP support

The **STP Topology Loop Predictor** is a lightweight and efficient solution for analyzing Layer-2 networks. It improves network reliability by predicting loops, validating STP configurations, and providing intelligent recommendations before deployment.